Yili Wen

Last updated: July 2026

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Research Interests

I'm interested in designing and building **wearable and robotic systems that restore and augment human movement**. I am drawn to the full loop from human intent to physical assistance, and I work across **actuation and control, embedded sensing, and digital fabrication**. My approach is hands-on and human-centered. I prototype real hardware and evaluate it with the people it is built for. Looking ahead, I want to extend my research toward **assistive and rehabilitation robotics and exoskeletons**, and toward closing the loop between neural intent and adaptive physical assistance.

Education

2025 – Present **Graduate Student, Computer Science — The University of Texas at Dallas**
Design & Engineering for Making (DE4M) Lab, advised by Liang He.
M.S. in Computer Science (Interactive Computing) expected **Spring 2027**.

2021 – 2025 **B.A. in Computation and Design — Duke Kunshan University**
Dual-degree program with **B.A. in Interdisciplinary Studies, Duke University**.
Undergraduate research advisor: Xin Tong.

Publications

2026 [E.1] ***Yili Wen**, Yafei Ge, Xin Tong, Liang He. "PufFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication." (Work-in-Progress). *ACM TEI 2026.*

2026 [W.2] ***Yili Wen**, Ceci Verbaarschot, Liang He. "From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients." (Workshop paper & poster). *Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026.*

2026 [P.1] Georgia Loewen, Karen Anne Cochrane, **Yili Wen**, Audrey Girouard. "Personas at Play: Understanding Disabled Player Experience to Expand Accessible Controller Design." (Poster). *Graphics Interface (GI) 2026.*

2025 [C.1] ***Yili Wen**, Jiaxin Wang, Hongni Ye, Liang He, Min Fan, Xin Tong. "'I Want to Use Technology to Help Farmers in the Future': Enhancing Children's Understanding of Farming through Tangible User Interfaces and Embodied Interaction." *ACM Interaction Design and Children (IDC) 2025.*

2024 [C.2] Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. "'I Keep Sweet Cats in Real Life, But What I Need in the Virtual World Is a Neurotic Dragon': Virtual Pet Design with Personality Patterns." *Chinese CHI 2024.*

2023 [W.1] Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. "PetGen: Design and Generation of Virtual Pets." (Workshop). *IEEE ICME Workshops (ICMEW) 2023.*

Manuscripts In Preparation

– ***Yili Wen**, Ceci Verbaarschot, Liang He. "A Soft Grasp-Assist Exoskeleton for Spinal Cord Injury: Design and Intent-Driven Control." In preparation; targeting *IEEE Robotics and Automation Letters (RA-L).*

– ***Yili Wen**, Liang He. "Sequential vs. Simultaneous Actuation for Guiding Blind and Low-Vision Runners." In preparation; targeting *ACM IMWUT* (Nov. 2026 cycle).

– Tianjian Liu, Jingwei Sun, **Yili Wen**, Haiqi Shen, Liuxin Zhang, Yu Zhang, Qianying Wang, Xin Tong. "Human–AI Co-Writing in Knowledge-Intensive Tasks: Behaviors, Challenges, and Collaboration Patterns." In preparation; targeting *Int. J. Human–Computer Interaction (IJHCI).*

Selected Research Projects

2025 – Present

Soft Grasp-Assist Exoskeleton for Spinal Cord Injury **("From Intention to Action")**

A wearable soft exoskeleton that restores grasping to people with spinal cord injury, driven by the wearer's own decoded intent.

- Designed soft pneumatic finger actuators (printed silicone using Form4B) and a layer-jamming wrist brace for stable, compliant grasp assistance during everyday tasks.
- Built the embedded control and pressure-sensing stack, and a neural-intent (BCI) pipeline that turns decoded intention into adaptive assistance.
- Ground the design in real rehabilitation needs through a co-design study with clinicians.

2025 – Present

Wearable Actuation System for Guiding Blind & Low-Vision Runners

A body-worn actuation system that gives blind and low-vision runners directional turn and hazard guidance on their own body, so they can run independently instead of relying on a sighted guide.

- Built distributed spinal actuator tiles (motorized press and lateral drag) that reproduce a guide's directional pull, paired with on-device vision and a companion iOS app.
- Investigating a novel **sequential (cranio-caudal) vs. simultaneous** actuation scheme, grounded in turning biomechanics and a formative study with BLV runners and guides.

2025

PufFab: Shape-Transformable Rice Paper for Playful Food Fabrication **(TEI '26)**

A design-to-fabrication pipeline that lets people make edible rice paper puff from flat patterns into raised 2.5D forms when fried.

- Modified a 3D printer and built the design tool so users customize a flat pattern, using moistening and cutting to control how each region expands or curls during frying.

2024 – 2025

Adaptive Game-Controller Hardware for Upper-Limb Motor Disabilities **(GI '26)**

DIY, reconfigurable game controllers that let players with upper-limb motor disabilities shape the input to their own body.

- Designed and fabricated the modular, remappable controller prototypes that served as the physical platform for the user study.
- Co-authored the resulting poster (Graphics Interface 2026), which studied player experience and narrative-inquiry methods around this platform.

2023 – 2024

WooGu: Tangible System for Children's Agricultural Literacy **(IDC '25)**

An educational game that teaches children farm-to-table knowledge through a tangible interface, sensor-embedded 3D-printed tools, and whole-body play.

- Designed and built the system, pairing sensor-embedded 3D-printed farming tools with an augmented digital display to give coordinated visual, tactile, and digital feedback.
- Evaluated with 15 children aged 5 to 12, where it improved their farming knowledge, engagement, and attitudes toward agriculture.

Research Experience

2025 – Present

Research Assistant — Liang He

Design & Engineering for Making (DE4M) Lab, UT Dallas, USA.

2023 – 2025

Research Assistant — Xin Tong

HCI Lab, Duke Kunshan University & Information Hub, HKUST (Guangzhou), China.

2023 – 2025

Research Assistant — Karen Cochrane

Embodied Computing Lab, University of Waterloo, Canada.

2022 – 2023

Research Assistant — Fan Liang

Center for the Study of Contemporary China, Duke Kunshan University, China.

Teaching & Outreach

Summer 2026

Instructor — TAST STEM Bridge Summer Camp

Department of Computer Science, UT Dallas. Six-week camp (June–July 2026); designed and delivered hands-on lectures and build sessions.

Summer 2026

Instructor — CS Outreach Summer Camp

Department of Computer Science, UT Dallas. June–July 2026.

Spring 2026	**Teaching Assistant — CS 4376 Object-Oriented Design** Department of Computer Science, UT Dallas.
Fall 2025	**Teaching Assistant — CS 6326 Introduction to Human–Computer Interaction** Department of Computer Science, UT Dallas.

Technical Skills

- **Fabrication & Prototyping:** soft-robotics fabrication (silicone molding/casting), SLA (resin) & FDM 3D printing, OpenSCAD / CAD.
- **Electronics & Embedded:** ESP32 / Arduino, soft pneumatic and DC-motor actuation, closed-loop sensor control, wireless (ESP-NOW).
- **Machine Learning & Vision:** custom object detection (YOLO / Ultralytics), on-device deployment (Core ML), OpenCV.
- **Programming:** Python, C/C++ (embedded), Swift/SwiftUI, JavaScript/p5.js, Java/Processing, R.
- **Design & Analysis:** Figma, Blender, statistical analysis in R.
- **Research Methods:** human-subjects studies (interviews, surveys, in-field evaluation), mixed methods, sensor-data analysis.

Presentations

Apr 2026	**Poster & Talk — "From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients"** Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026, Barcelona, Spain.
2026	**Talk & Poster — "PufFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication"** ACM TEI 2026.
2025	**Paper Presentation — " 'I Want to Use Technology to Help Farmers in the Future' "** ACM Interaction Design and Children (IDC) 2025.

Awards

2023	**Center for the Study of Contemporary China (CSCC) Fellowship** Duke Kunshan University.
2022 – 2024	**Dean's List** Duke Kunshan University (Fall 2022, Fall 2023, Spring 2024, Fall 2024).